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Hard-Budget Feature Selection with Resampled Mutual-Information Repair in EHOA

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Abstract. We compare four budget repairs within the Enhanced Hiking Optimization Algorithm (EHOA) while preserving its search and wrapper objective. The proposed repair ranks features by a nonnegative mean-minus-standard-deviation mutual-information score divided by synthetic acquisition cost. This score is a conservative heuristic, not a calibrated confidence bound. We audit 300 archived runs on ten classification datasets. All 240 constrained outputs satisfy a budget of 30% of total feature cost. Source inspection identifies target leakage in Steel Plates and a subject-overlap risk in Parkinsons; we exclude both from the main retrospective analysis. Across eight eligible datasets, resampled repair changes mean balanced accuracy from 0.8068 to 0.8116, cost ratio from 0.4791 to 0.2792, and Nogueira stability from 0.1161 to 0.2550. Friedman tests detect method differences in cost and stability, but not balanced accuracy. The resampled-versus-EHOA stability contrast does not survive within-metric Holm correction. A breast-cancer cytology case illustrates subset selection without claiming clinical validation or monetary savings. The contribution is an auditable repair comparison, not general predictive superiority. A separate 60-run exploratory correction also finds no general predictive advantage.

Keywords. feature selection, hard budget, resampled mutual information, subset stability

1 Problem and contribution

A small feature set need not be inexpensive: acquisition costs can differ between variables. Hard-budget selection limits the sum of selected costs [1]. Subset stability measures how feature choices vary across training samples and random search, rather than how accuracy or convergence varies [2]. In biomedical classification, this distinction matters when investigators interpret a selected measurement panel.

Medical feature selection spans imaging, signals and molecular data [3]. Predictive discrimination, reproducible feature identity and acquisition burden are separate objectives. Correlated descriptors can substitute for one another, so a changing mask need not imply poor prediction; conversely, a repeated mask does not establish a valid biomarker. This motivates reporting balanced accuracy, cost and chance-corrected subset stability together.

We retain the implemented EHOA search [4] and compare unconstrained EHOA with four repairs. Only the repair utility changes among constrained variants. The contribution is the integration and controlled evaluation of resampled relevance-per-cost repair. We do not claim that budgets, repair, mutual information (MI), or resampling are new. CostStableEHOA remains the code identifier; the revised title avoids implying stable costs or proven selection stability.

2 Related studies

Stability metrics can reward reproducible yet weak predictors; importance-weighted stability addresses this concern [5]. Medical budget selection with penalized risk [6], sequential acquisition [7], and grouped costs with shadow features [8] solve related but different problems. Table 1 summarizes these distinctions; NR means not reported in the inspected source. Their numerical results are not directly comparable with our protocol.

Table 1. Related-work comparison. Full evaluation details are in the supplement.

Study	Budget	Costs	Stability	Evaluation	Difference
Jagdhuber [1]	Yes	Additive	NR	Simulation	Feasible GA
Nogueira [2]	NR	NR	Chance-corrected	Resampling	Metric
Hamer [5]	NR	NR	Importance-weighted	Resampling	Metric limits
Klonecki [6]	Yes	Feature / proxy	NR	Medical data	Penalized ERM
Valancius [7]	NR	Sequential	NR	AFA benchmarks	Patient-wise policy
Klonecki [8]	Yes	Grouped	NR	MIMIC	Shadow stopping
This work	Yes	Synthetic additive	Nogueira	2 x 3 outer	MI/cost repair

Classical stability selection aggregates repeated subsample selections and, under specified assumptions and thresholds, controls false selections [9,10]. Our four MI resamples instead estimate a deletion utility before wrapper search. We neither threshold repeated selection probabilities nor inherit those error-control guarantees. Nogueira stability evaluates the final masks [2]; it is not the optimized fitness. Active acquisition [7] chooses measurements sequentially per case, whereas our repair returns a fixed feasible subset.

3 Implemented EHOA

Each hiker has a continuous position x and a velocity v; a binary mask z selects features. Initialization uses a tent sequence with initial value 0.7 and parameter 0.4, mapped to [-6,6]. Velocities start uniformly in [-1.2,1.2]. For iteration t of T=10, the implemented update is:

$$v_i \leftarrow clip[-12,12](w_t v_i + SF_t[2r_1(p_i - x_i) + 2r_2(g - x_i)])$$
$$x_i \leftarrow clip[-6,6](x_i + v_i)$$
$$w_t = 0.9 - 0.5t/T;\ SF_t = 3 - 2t/T$$

Here p_i and g are personal and global best continuous positions, and r_1,r_2 are independent per-feature uniform draws. The sigmoid gives selection probability 1/(1+exp(-x_j)); an empty sampled mask restores the largest probability. The implementation keeps continuous leaders separate from repaired masks, caches fitness for duplicate masks, and stops after ten iterations without early stopping.

4 Budget repair and stability

Costs are fixed positive additive synthetic units, independent of labels: c is drawn once per dataset from lognormal(0,0.7), with NumPy seed 1000+D. The absolute budget B=0.30 sum(c) is derived from a relative budget. Ionosphere and Dermatology share a cost vector because both have D=34. A nonempty mask is feasible when sum(z_j c_j) <= B.

$$u_j = max(0, \bar{I}_j - \lambda s_j);\ q_j = u_j/c_j;\ \lambda = 1$$

For resampled repair, four within-class samples draw ceil(0.8 n_class) rows with replacement (minimum two). Dense mutual_info_classif uses continuous-feature estimation, three neighbors, and nats; missing values are zero-filled for MI only. s_j is the sample standard deviation (ddof=1). No division by sqrt(R) or calibrated confidence level occurs. Utility is computed once on the outer-training partition using a separate RNG seeded by selector_seed+7919.

Table 2. Four repairs share the same deletion rule.

Repair	Utility u_j	Meaning
Random/cost	U(0,1), fixed per fit	Not uniform deletion
Cheapest	1	Largest cost first
MI/cost	Single-sample MI	Relevance per cost
Resampled/cost	max(0, mean MI - SD)	Conservative heuristic

Procedure: initialize population; sample sigmoid mask; repair before fitness; evaluate; update personal/global leaders; repeat movement, sampling, repair, and evaluation; return the best repaired mask. Repair removes argmin(q_j) among active features while cost exceeds B and more than one feature remains. An unaffordable singleton is cleared; an empty mask restores the affordable feature maximizing q_j. A tie selects the first feature index. Utility is never recomputed per candidate.

$$F(z) = 0.99(1 - Ac\bar{c}inner) + 0.01|z|/D$$

The unchanged objective minimizes ordinary inner accuracy error, not balanced-accuracy error. External balanced accuracy (BA) is the mean recall across classes. Stability is computed once per dataset/method from M=6 final masks:

$$s^2_j = M/(M-1) \cdot p_j(1-p_j)$$
$$\Phi = clip[-1,1](1 - mean(s^2_j)/[(\bar{k}/D)(1-\bar{k}/D)])$$

p_j is selection frequency and k-bar is mean subset size. Identical empty/full masks return one in the code. This dimensionless chance-corrected outcome [2] mixes data-split and search variation; overlapping training folds make the six masks dependent. We do not treat copied stability values in six CSV rows as six observations. Jaccard overlap is supplementary and is not chance-corrected.

The continuous-discrete nearest-neighbor MI estimator [11] measures marginal dependence, not conditional contribution after other features are selected. Univariate ranking can retain redundant correlated variables or discard features useful only jointly. With four resamples, the estimated spread is imprecise; lambda=1 expresses a fixed penalty rather than a coverage probability. These design limits motivate comparison with the simpler single-MI and cheapest repairs, rather than attributing every difference to stability.

Feasibility follows from positive costs and finite deletion: every deletion reduces both subset size and total cost. An unaffordable last feature is cleared; restoration chooses only an individually affordable feature. Thus a nonempty feasible output exists when at least one feature costs at most B, as verified for these datasets. This proves budget compliance, not optimal utility or accuracy. Repair may remove a jointly useful feature because its ranking is marginal and fixed. Repeated repair also maps different search positions to the same mask; cached fitness avoids evaluating such duplicates again.

5 Datasets and source audit

Table 3. Snapshot / deduplicated N; D is the supplied numeric feature count.

Dataset	N raw/used	D	K	Class counts used	Source
Breast Cancer	569 / 569	30	2	212/ 357	U17
Wine	178 / 178	13	3	59/ 71/ 48	U109
Ionosphere	351 / 350	34	2	125/ 225	O59
Sonar	208 / 208	60	2	111/ 97	O40
Dermatology	366 / 366	34	6	112/ 61/ 72/ 49/ 52/ 20	O35
Spambase	4601 / 4210	57	2	2531/ 1679	O44
Parkinsons*	195 / 195	22	2	48/ 147	O1488
Hill-Valley	1212 / 1212	100	2	606/ 606	O1479
Steel Plates*	1941 / 1941	33	2	1268/ 673	O1504
Madelon	2600 / 2600	500	2	1300/ 1300	O1485

U identifiers link to archive.ics.uci.edu/dataset/{id}; O identifiers link to openml.org/d/{id}. The supplement gives clickable direct links, raw/used class counts, hashes and budgets. No runtime one-hot encoding occurs. Steel has 33 inputs and two classes in the snapshot, whereas UCI defines 27 predictors and seven fault types. Its V28-V33 are six fault indicators and exactly determine the target; we exclude these results. Parkinsons is a recording-level dataset with repeated participants and no subject-group split; we also exclude it from the main analysis. Eight eligible datasets retain 240 runs.

6 Evaluation and statistical analysis

SMOTE creates synthetic minority-class training examples [12]; it does not increase the number of independent test subjects. Model selection must remain separate from performance estimation [13]. Here the inner score guides search, while untouched outer tests provide the reported predictions. MI sees the full outer-training labels, including inner-validation labels; thus the inner score is not an unbiased validation estimate. The corrected subject splits address recording overlap, but do not change this utility design.

Table 4 documents actual settings rather than class defaults. All methods share data splits, costs, seeds, search equations and wrapper fitness. Initialization positions are deterministic; search velocities and masks use the selector RNG. The four repairs therefore isolate utility ranking under a common budget; unconstrained EHOA additionally tests the effect of imposing that budget.

Table 4. Settings extracted from code and configuration.

Parameter	Value
Population / iterations	8 / 10; no early stop
Position / velocity clip	[-6,6] / [-12,12]
Tent / sweep / inertia	0.7, p=0.4 / 3 to 1 / 0.9 to 0.4
c1,c2 / alpha	2,2 / 0.99
Outer / inner folds	3 stratified / 3 stratified
Outer seeds	71001, 71002
Selector seed	outer seed + fold index x 100003
MI resamples / fraction	4 / 0.8 with replacement
lambda / MI neighbors	1 / 3
Cost / budget	lognormal(0,0.7) / 30% total
Classifier	5-NN; uniform; Euclidean
Imputation / scale	Training mean / StandardScaler
SMOTE	Training only; k=min(5,n_min-1)

Within each inner fold, training data alone fit mean imputation, scaling and SMOTE; validation rows remain unchanged. MI utility uses all outer-training labels, including rows that later become inner validation. Thus inner scores are selection scores, not unbiased estimates. Outer-test labels do not enter selection. Final outer training fits preprocessing/SMOTE and 5-NN; archived metrics retain only 5-NN although the runner fits four classifiers. Preprocessing and SMOTE occur before masking; this is a simulation, not a deployed selected-only acquisition pipeline.

We average six BA/cost/size values per dataset and use one stability value per method/dataset. Eight datasets, not 240 rows, are analysis blocks [14]. The 2026-10-12 analysis is retrospective: the supplied confirmatory flag and config hash do not establish dated preregistration. We use tie-corrected Friedman tests (df=4), two-sided paired Wilcoxon contrasts of resampled repair versus four controls, and Holm correction within each metric. The supplement also corrects all 24 contrasts and six omnibus tests. Dataset-bootstrap intervals (20,000 draws; seed 20261012) are descriptive, unadjusted percentile intervals.

7 Results

Table 5. Equal-weight means over eight eligible datasets; V is final budget exceedance.

Method	BA	Cost ratio	Size	Stability	V
EHOA	0.8068	0.4791	49.21	0.1161	47/48
Random/cost	0.8054	0.2727	37.54	0.1238	0/48
Cheapest	0.8090	0.2708	41.50	0.2423	0/48
MI/cost	0.8109	0.2770	32.65	0.2287	0/48
Resamp./cost	0.8116	0.2792	36.40	0.2550	0/48

Table 6. Friedman statistic and raw p; average ranks in E/R/C/M/S order, 1 is best.

Metric	Chi2	p	E	R	C	M	S
BA	4.5000	0.34255	3.12	3.88	3.00	2.75	2.25
Cost	21.1000	0.00030	5.00	1.88	1.88	3.00	3.25
Size	23.7233	0.00009	5.00	1.88	3.81	2.06	2.25
Stability	15.7000	0.00345	4.25	4.12	1.88	2.50	2.25

Resampled repair versus EHOA changes cost ratio by -0.19986 (Wilcoxon W=0, raw p=0.00781; within-metric Holm p=0.03125; rank-biserial r=-1). Stability changes by +0.13895 (W=1, raw p=0.01563; Holm p=0.06250; r=0.944). BA changes by +0.00482 (W=9; Holm p=0.750; r=0.500; 95% interval [-0.00149,0.01109]). Neither equivalence nor non-inferiority follows. No resampled repair contrast with the other three repairs is significant within metric for BA or Nogueira stability.

As a separate exploratory sensitivity analysis, we completed 60 corrected runs: Steel used 27 predictors and seven classes; Parkinsons used subject-disjoint outer and inner folds after matching all 195 recordings to the UCI file. Combining these with the 240 eligible archived runs gives ten corrected dataset blocks. Friedman BA remains nonsignificant (Chi2=4.32, df=4, p=0.36443). Resampled-versus-EHOA stability has within-metric Holm p=0.01563, but Holm over all 20 contrasts gives p=0.06641. Mean BA delta is -0.00583, with descriptive interval [-0.02523,0.00771]. These new exploratory results neither validate the original leaked experiment nor establish superiority over simpler repairs.

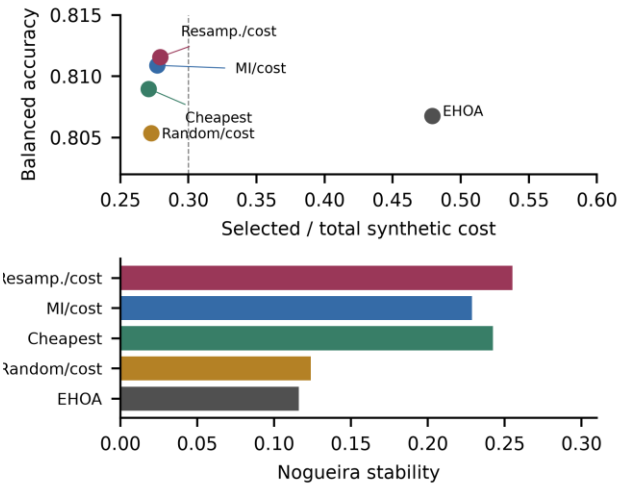


Figure 1. Eight-dataset means; cost uses synthetic units. Stability bars do not imply significant pairwise differences.

8 Biomedical case and congress relevance

The UCI diagnostic breast-cancer dataset contains 569 fine-needle-aspirate cases (212 malignant, 357 benign) and 30 nuclear-image descriptors [15]. Radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension each have mean, error and worst summaries. Table 7 illustrates the five methods on this dataset.

Table 7. Breast-cancer case, means over six runs.

Method	BA	Cost ratio	Size	Stability
EHOA	0.9563	0.5227	14.17	0.0682
Random/cost	0.9452	0.2651	10.17	0.0627
Cheapest	0.9557	0.2673	11.33	0.2815
MI/cost	0.9462	0.2758	10.50	0.2137
Resamp./cost	0.9525	0.2799	11.17	0.2059

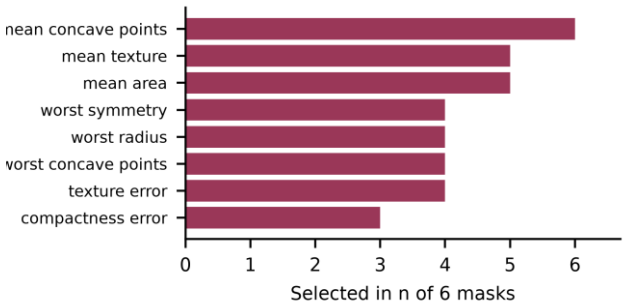


Figure 2. Selection counts for the eight most frequent resampled-repair features, from six archived masks.

Resampled repair selects mean concave points in 6/6 masks and mean texture and mean area in 5/6. These counts describe the archived selections; they do not establish causal biomarkers or clinical reliability. Cheapest repair has greater stability here (0.2815 versus 0.2059). Synthetic feature costs are not biopsy prices; all descriptors derive from the same acquisition, so grouped measurement costs would be more realistic.

The closest official BAIC2026 topic is Artificial Intelligence, Machine Learning and Data Science / Intelligent Disease Diagnosis Systems (baic.um.ac.ir/fa/home/Content/12). Our biomedical contribution concerns computational measurement-panel selection, not a clinically validated diagnostic system. We claim no cognitive-science mechanism.

Across the eight datasets, the proposed repair reduces mean synthetic cost by 41.7% and subset size by 26.0% relative to unconstrained EHOA. These comparisons include imposing a

budget on an otherwise infeasible baseline: 47 of its 48 outputs exceed that budget. They cannot isolate a benefit of resampling. The cheapest repair already attains mean cost 0.2708 and stability 0.2423, compared with 0.2792 and 0.2550 for resampled repair. The small descriptive stability difference is not corrected pairwise evidence. A medical panel decision therefore needs measured acquisition groups and acceptable sensitivity, rather than choosing the method from stability alone.

9 Limitations and reproducibility

Only one synthetic budget, four MI resamples, eight eligible datasets, two seeds and a short search were assessed. Ordinal Dermatology inputs are treated as continuous by MI; duplicated bootstrap rows can distort nearest-neighbor MI. Overlap-based stability does not verify clinical importance. The corrected exploratory follow-up is reported separately. UCI describes 31 Parkinson participants, but its file contains 32 distinct subject prefixes; group identity needs independent confirmation. Dated registration, original execution logs and end-to-end timing are unavailable; runtime_seconds excludes MI utility construction in constrained methods.

The accompanying local supplement contains all dataset links, 300 mask audits, per-dataset outcomes, cost vectors, statistics and regeneration scripts. Raw files and originals remain intact. No verified public code repository is supplied, and local availability is not public release. The eight-dataset exclusion and analysis are retrospective, not new confirmatory experiments.

10 Conclusion

All archived constrained outputs are budget-feasible. On eight eligible datasets, resampled repair shows a descriptive cost-stability trade-off but no corrected pairwise stability advantage or general predictive superiority. A stronger biomedical claim requires grouped real costs, subject-safe evaluation and external clinical validation.

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